De La Salle Holy Cross College

NOVEMBER EXAMINATION

2024

Grade 10

PHYSICAL SCIENCES

Paper 1: Physics

Examiner:	Mrs A. Kamedza	**Time:**	2 hours
Moderator:	Mr E. Hove	**Marks:**	130

MEMORANDUM

QUESTION 1 MULTIPLE CHOICE

Answer the multiple-choice questions on the answer grid below. Make a clear cross (X) in the box corresponding to the letter that you consider to be correct. Every question has only one correct answer.

A	B	C	D

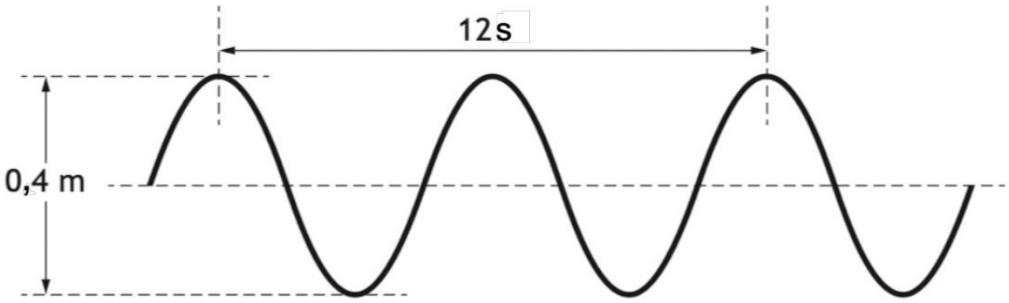
Here the option C has been marked as an example.

	A	B	C	D
1.1				X
1.2			X	
1.3	X			
1.4			X	
1.5		X		
1.6				X
1.7			X	
1.8		X		
1.9		X		
1.10	X			

[20]

QUESTION 2

2.1 The diagram below represents a water wave. The speed of the wave is 3,0 $m \cdot s^{-1}$.



2.1.1 What is type of wave is a water wave? (1)

Transverse wave✓

2.1.2 Define *frequency*. (2)

The number of complete waves that pass a given/fixed point in a second, ✓✓

2.1.3 Write down the amplitude and frequency for this wave: (2)

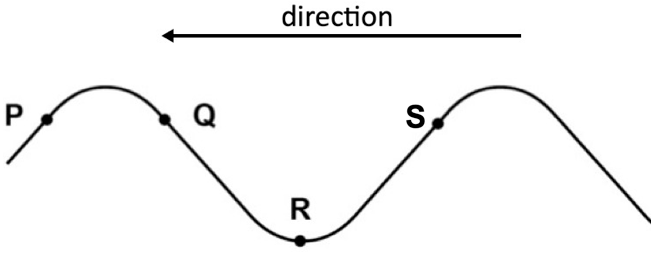
Amplitude: A = 0,2 m ✓

Frequency: f = $\frac{2}{12}$ = 0,17 Hz ✓

2.1.4 Hence, calculate the wavelength of this wave. (3)

v = fλ ✓ ∴ $\lambda = \frac{3}{0,17}$ ✓ = 17,65 m ✓ (18 m without rounding)

2.2 The diagram below shows a wave in a string at one instant in time.



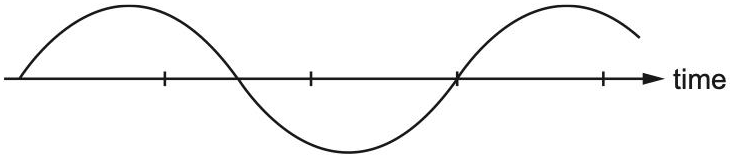
2.2.1 Identify two points on the wave that are in phase. (1)

P and S ✓

2.2.2 Identify two points on the wave that are out of phase. (1)

P and Q/ Q and R/ R and S ✓

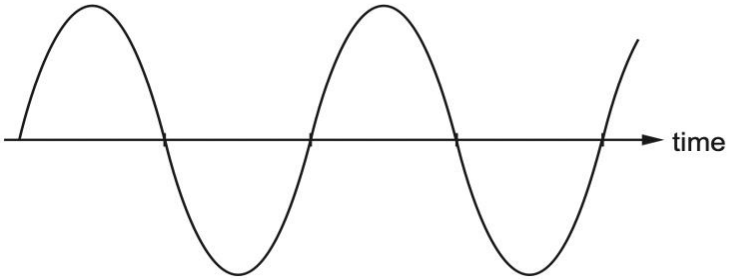
2.3 Consider the following diagram of a sound wave.



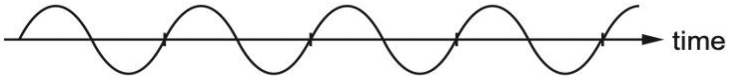
The diagrams below represent the waves produced by three other sources of sound. The scales are the same for all the diagrams.
Which wave represents a sound of higher pitch but lower volume than the original wave shown above?

Mark your choice by placing an X in the box. (2)

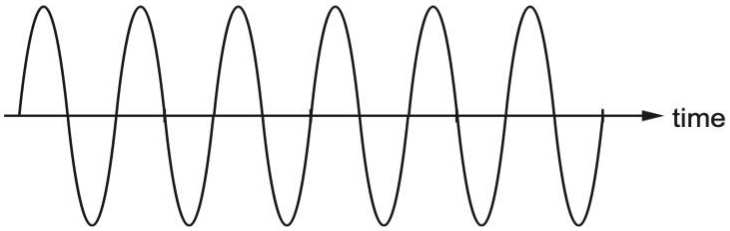
☐



☒ ✓✓



☐



[12]

QUESTION 3

3.1 What is meant by *electromagnetic radiation*? (3)

Accelerating charged✓ particles produce alternating/changing electric fields and magnetic fields✓ which are perpendicular✓ to each other (through space at the speed of light).

3.2 The diagram below shows regions of the electromagnetic (EM) spectrum.

radio waves	microwaves	infra-red	visible light	ultraviolet	X-rays	gamma rays

Identify **three** correct statements by placing an X in the corresponding boxes. (3)

All EM waves travel at the same speed in a vacuum.	X
The energy of EM waves decreases from radio waves to gamma waves.	
Microwaves, ultraviolet and X-rays are all ionizing radiation.	
The wavelength of EM waves decreases from radio waves to gamma rays.	X
Gamma rays can be emitted from radioactive materials.	X
All EM waves are mechanical and longitudinal.	

3.2 Visible light with wavelength of 7,8 nm is shone onto the surface of zinc and electrons are ejected.

3.2.1 What is evidenced by the observation above? (1)

The particle nature of light✓

3.2.2 What are the particles of light called and what do w know about their energy? (2)

Photons✓
They carry a discrete/specific amount of energy/frequency✓.

3.2.3 Calculate the energy of this light. (4)

$E = \frac{hc}{\lambda}$ ✓ $= \frac{(6{,}6\times10^{-34})(3\times10^{8})✓}{7{,}8\times10^{-9}✓(\text{conversion})}$ = 2,54 x 10^{-17} J ✓

[13]

QUESTION 4

4.1 MRI machines employ powerful magnets which produce a strong magnetic field that forces protons in the body to align with the field.

4.1.1 What is meant by the term *magnetic field*? (2)

A region in space/area around a magnet✓ where another magnet/ferromagnetic material will experience a force✓.

4.1.2 Explain how an electric current can turn an iron nail into a magnet. (3)



[Image source: https://carnegiestemgirls.org/wp-content/uploads/2016/04/nail-electromagnet-300x254.jpg]

- A current carrying wire has a magnetic field around it. ✓
- An iron nail has magnetic domains✓
- The iron nail's domains align✓ due to the field around the wire making it a temporary magnet.

4.2 To identical spheres are used in an experiment to determine the relationship between the force between them and the magnitude of the charge.
The results of the experiment are given below.

4.2.1 State the variables below: (4)

Independent Variable: Charge on sphere A ✓

Dependent variable: Force on sphere B ✓

2 Controlled variables: Distance between spheres✓; Charge on sphere B✓

4.2.2 Write down the hypothesis for the investigation. (2)

The force on sphere B is directly proportional to the charge on sphere A. ✓✓

4.2.3 Plot a graph of the force on sphere B versus the charge on sphere A. (6)

Heading✓
Y-axis label + unit (Force) ✓
X-axis label + unit (Charge) ✓
Plotting ✓✓
LOBF through origin ✓

4.2.4 State the relationship between the force and the charge on the sphere. (2)

The force on sphere B is directly proportional to the charge on sphere A. ✓✓
The greater the charge the higher the force✓

[19]

QUESTION 5

Use the circuit diagram below to answer the questions that follow:

5.1 Define the following terms:

5.1.1 *current strength*. (2)

The rate of flow of charge✓✓

5.1.2 *Ohm's law*. (2)

The current through a conductor is directly proportional to the potential difference across a conductor at a constant temperature. ✓✓

5.2 Calculate the current reading on ammeter A_2. (3)

$I = \frac{V}{R} = \frac{14.4✓}{30✓} = 0,48$ A ✓

5.3 Determine the resistance of resistor X. (5)

$I_x = 0,72 – 0,48$ ✓ $= 0,24$ A ✓

$R_x = \frac{V}{I} = \frac{14.4✓}{0,24✓} = 60\ \Omega$ ✓

5.4 State the reading on voltmeter V_3. (2)

$V_3 = I \times R = 0,72 \times 80$ ✓ $= 57,6$ V ✓

5.5 Determine the total voltage (V_{TOTAL}) across the battery. (2)

$V_{TOTAL} = 14,4 + 57,6$ ✓ $= 72$ V ✓

OR

$V_{TOTAL} = I_T R_T = 0,72 \times 100$ ✓ $= 72$ V ✓

5.6 Calculate the charge that flows past point Z in 2,5 minutes. (4)

$Q = I \times t$ ✓$= 0,72$ ✓ $\times (2,5 \times 60)$ ✓ $= 108$ C ✓

[20]

QUESTION 6

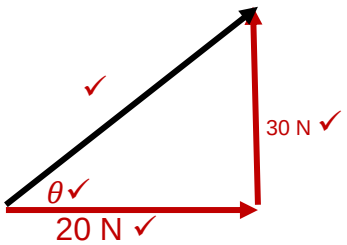
A wooden block is pushed with a force of 30 N North and 20 N East.

6.1 Define a *vector quantity*. (2)

A physical quantity with both magnitude and direction. ✓✓

6.2 Use a vector diagram drawn to scale or algebraic methods to determine the resultant force. (5)

Diagram method:



F_{res} = 36,06 N at 34 ^{o} ✓

Algebraic method:

$F_{res} = \sqrt{(20^2 + 30^2)}$ ✓ = 36,06 N ✓

$\tan \theta = \frac{30}{20}$ ✓ ∴ θ = 33,7 ^{o} ✓

F_{res} = 36,06 N at 33,7^{o} ✓

6.3 Thus, write down the equilibrant force using your answer to Question 6.2. (2)

36,06 N✓ at 213,7^{o} ✓

[9]

QUESTION 7

A car is initially travelling **east** along a straight road. A velocity vs. time graph of its motion is given below.

7.1 Define *displacement*. (1)

The change in position✓ of an object.

7.2 At point **G**, after 30 s, is the car's displacement east or west? Explain your answer. (3)

West ✓
The area under the graph represents displacement✓
The area below the x-axis is greater than the area above✓
∴ net displacement in west

7.3 Calculate the car's average velocity for the first 15 seconds. (6)

Δx = Area under the graph = (½x5x30) ✓+ (½x5x-30) ✓+(5x-30) ✓= -150 m

$v_{average} = \frac{\Delta x}{\Delta t} = \frac{-150}{15}$ ✓ = -10 m/s✓ = 10 m/s West ✓

7.4 During which interval(s) (A - G) is the car:

7.4.1 travelling east? A-B/FG ✓ (1)

7.4.2 experiencing constant negative acceleration? AB/AC/BC ✓ (1)

7.4.3 stationary for more than one second? EF ✓ (1)

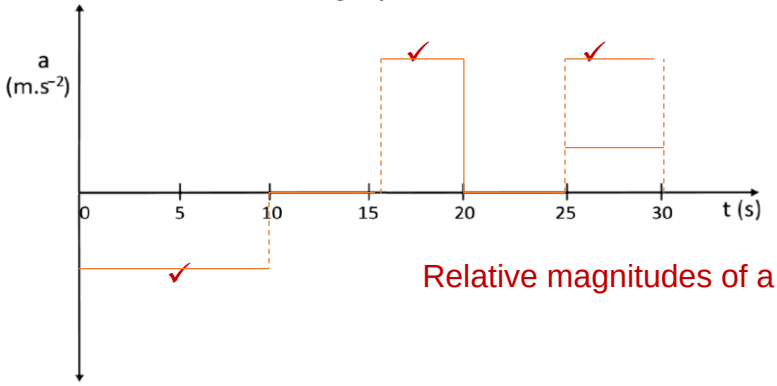
7.5 Define *acceleration*. (2)

The rate of change of velocity✓✓

7.6 Calculate the car's acceleration between 15 s and 20 s. (4)

$a = \frac{\Delta v}{\Delta t} = \frac{0-(-30)✓}{20-15✓}$ = 6 m·s^{-2}✓ East ✓

7.7 Hence, draw an acceleration vs. time graph for the 30 second period. (3)
No values are needed but the graph must correlate to the time values on the x-axis.



Relative magnitudes of a need to be correct

[22]

QUESTION 8

Our entire universe is governed by energy. It manifests in many different forms, but its origins remain a mystery.

8.1 Define *gravitational potential energy*. (2)

Energy due to an object's position✓ relative to a reference point✓.

8.2 Consider the following two situations:

1. A 270 kg gorilla runs at you at 5 $m \cdot s^{-1}$.
2. A 25 kg badger runs at you at 25 $m \cdot s^{-1}$.

Which situation indicates the animal with the most kinetic energy?
Use calculations to explain your answer. (5)

1. $E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 270 \times 5^2$ ✓ = 3 375 J ✓

2. $E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 25 \times 25^2$ ✓ = 7 812,5 J ✓

∴ The badger has more kinetic energy✓

8.3 State the *principle of conservation of mechanical energy*. (2)

In the absence of air resistance, or any external forces, the mechanical energy of an object is constant. ✓✓
OR
The mechanical energy of an isolated system remains constant. ✓✓

8.4 A skier, including her equipment, with a total mass of 80 kg, approaches the edge of a ski slope 50 m high at a speed of 10 $m \cdot s^{-1}$ as shown in the diagram below.
Ignore air and surface friction.
Calculate the speed of the skier at the bottom of the slope. (6)

$$E_{MA} = E_{MB}$$
$$E_{KA} + E_{pB} = EkB + E_{pB} \checkmark$$
$$(\frac{1}{2} \times 80 \times 10^2) \checkmark + (80 \times 9{,}8 \times 50) \checkmark = (\frac{1}{2} \times 80)\, v^2 \checkmark + 0 \checkmark$$
$$39\,200 + 4\,000 + 40\, v^2$$
$$v = 32{,}86\ m \cdot s^{-1} \checkmark$$

[15]

Total: 130 marks